

Advanced DAX Measures

Practical No.:	PL-300 Practical 5
Practical Title:	Create Advanced DAX Calculations in Power BI Desktop
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Project Summary

I built a tabular matrix and manipulated the filter context using CALCULATE() to develop regional, country, and group sales percentages. Additionally, I applied time intelligence functions, specifically TOTALYTD, to compute fiscal Year-to-Date sales and Year-over-Year growth, organizing these new measures into dedicated ratio and time intelligence metric groups.

Objectives

<i>Use the CALCULATE () function to manipulate filter context</i>	<i>Use Time Intelligence functions</i>
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Overview

1. Created a matrix visual and changed the layout to Tabular
2. Manipulated the filter context by
 - a. Adding the Sales All Region measure then converted it to percentages for readability,
 - b. Added the Sales % Country and improved it
 - a. Added the Sales % Group and improved it
3. Created a YTD measure
 - a. Added a Sales YTD measure
Sales YTD = TOTALYTD(SUM(Sales[Sales]), 'Date'[Date], "6-30")
4. Created a YoY growth measure
5. Created groups:
 - a. Time Intelligence
 - b. Ratios

Notes

Rands were used as currency instead of USD; hence the results will differ in currency but still be the same in terms of values

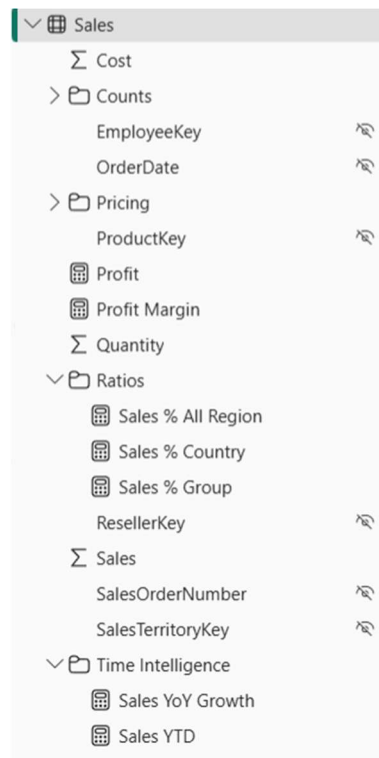
Images

Group	Country	Region	Sum of Sales	Sales % All Region	Sales % Country	Sales % Group
Europe	France	France	\$4 527 840	5,84%	100,00%	44,01%
		Total	\$4 527 840	5,84%		44,01%
	Germany	Germany	\$1 877 743	2,42%	100,00%	18,25%
		Total	\$1 877 743	2,42%		18,25%
	United Kingdom	United Kingdom	\$3 883 043	5,01%	100,00%	37,74%
		Total	\$3 883 043	5,01%		37,74%
	Total		\$10 288 626	13,27%		
North America	Canada	Canada	\$13 875 633	17,89%	100,00%	21,07%
		Total	\$13 875 633	17,89%		21,07%
	United States	Central	\$7 633 387	9,84%	14,68%	11,59%
		Northeast	\$6 715 354	8,66%	12,92%	10,20%
		Northwest	\$12 004 822	15,48%	23,09%	18,23%
		Southeast	\$7 638 607	9,85%	14,69%	11,60%
		Southwest	\$18 001 116	23,21%	34,62%	27,33%
		Total	\$51 993 286	67,05%		78,93%
	Total		\$65 868 919	84,94%		
Pacific	Australia	Australia	\$1 391 025	1,79%	100,00%	100,00%
		Total	\$1 391 025	1,79%		100,00%
	Total		\$1 391 025	1,79%		
Total			\$77 548 570	100,00%		

Figure 1: Summary of matrix visual based on the manipulated filter context

Year	Avg Price	Median Price	Min Price	Max Price	Orders	Order Lines	Sum of Sales	Sales YTD	Sales YoY Growth
FY2018	\$748,6778	\$419,46	\$4,75	\$2 146,96	739	8459	\$16 429 043	\$16 429 042,6	
2017 Jul	\$655,5871	\$419,46	\$5,19	\$2 146,96	38	352	\$489 328	\$489 328,4	
2017 Aug	\$738,9349	\$419,46	\$4,75	\$2 146,96	75	785	\$1 540 072	\$1 540 072,2	
2017 Sep	\$741,8502	\$419,46	\$5,19	\$2 146,96	60	593	\$1 166 332	\$1 166 332,2	
2017 Oct	\$677,4529	\$419,46	\$5,19	\$2 146,96	40	499	\$844 833	\$844 833,2	
2017 Nov	\$752,3056	\$419,46	\$5,01	\$2 146,96	90	1106	\$2 325 755	\$2 325 755,2	
2017 Dec	\$734,5822	\$419,46	\$5,01	\$2 146,96	63	803	\$1 703 435	\$1 703 435,2	
2018 Jan	\$808,9406	\$419,46	\$5,19	\$2 146,96	40	377	\$713 230	\$713 230,2	
2018 Feb	\$896,7953	\$419,46	\$5,01	\$2 146,96	79	866	\$1 900 794	\$1 900 794,2	
2018 Mar	\$863,5415	\$419,46	\$5,19	\$2 146,96	64	653	\$1 455 280	\$1 455 280,2	
2018 Apr	\$732,2498	\$419,46	\$5,19	\$2 146,96	37	494	\$883 011	\$883 011,2	
2018 May	\$761,3017	\$419,46	\$4,75	\$2 146,96	85	1112	\$2 269 720	\$2 269 720,2	
2018 Jun	\$552,9459	\$419,46	\$4,75	\$2 146,96	68	819	\$1 137 252	\$1 137 252,2	
FY2019	\$397,8098	\$202,33	\$4,32	\$1 466,01	1255	21670	\$27 979 780	\$27 979 779,53	70,31%
2018 Jul	\$350,7395	\$196,33	\$4,75	\$1 466,01	72	1723	\$2 411 559	\$2 411 558,5	392,83%
2018 Aug	\$372,7469	\$202,33	\$4,75	\$1 466,01	139	2964	\$3 615 914	\$3 615 914,2	134,79%
2018 Sep	\$365,2962	\$196,33	\$4,32	\$1 466,01	111	2185	\$2 804 647	\$2 804 647,2	148,18%
2018 Oct	\$381,9956	\$202,33	\$5,19	\$1 466,01	73	1406	\$1 804 177	\$1 804 177,2	113,55%
2018 Nov	\$412,479	\$209,26	\$5,01	\$1 466,01	133	2345	\$3 054 997	\$3 054 997,2	31,36%
2018 Dec	\$407,0772	\$202,33	\$4,75	\$1 466,01	114	1732	\$2 188 206	\$2 188 206,2	28,46%
2019 Jan	\$428,8728	\$324,45	\$5,19	\$1 466,01	65	983	\$1 318 592	\$1 318 592,2	84,88%
2019 Feb	\$488,754	\$469,79	\$5,19	\$1 466,01	132	1664	\$2 386 073	\$2 386 073,2	25,53%
2019 Mar	\$450,0264	\$324,45	\$5,01	\$1 466,01	106	1215	\$1 564 295	\$1 564 295,2	7,49%
2019 Apr	\$369,778	\$198,04	\$5,01	\$1 466,01	74	1449	\$1 868 433	\$1 868 433,2	111,60%
2019 May	\$400,5136	\$202,33	\$4,75	\$1 466,01	134	2306	\$2 882 638	\$2 882 638,2	27,00%
2019 Jun	\$390,3237	\$202,33	\$4,75	\$1 466,01	102	1698	\$1 990 249	\$1 990 249,2	75,01%
FY2020	\$392,1224	\$200,05	\$1,33	\$1 466,01	1622	27722	\$33 139 748	\$33 139 748,07	18,44%
2019 Jul	\$350,5536	\$158,43	\$1,37	\$1 466,01	94	2186	\$2 729 167	\$2 729 167,2	13,17%
2019 Aug	\$342,1944	\$153,89	\$1,33	\$1 466,01	185	3754	\$4 306 549	\$4 306 549,2	19,10%
2019 Sep	\$343,4048	\$158,43	\$1,37	\$1 466,01	176	3760	\$4 153 399	\$4 153 399,2	43,49%
2019 Oct	\$378,6459	\$158,43	\$1,37	\$1 466,01	99	1810	\$2 293 200	\$2 293 200,2	27,11%
2019 Nov	\$395,8025	\$200,05	\$1,37	\$1 466,01	178	2904	\$3 490 438	\$3 490 438,2	14,25%
2019 Dec	\$412,5221	\$202,33	\$1,37	\$1 466,01	176	3029	\$3 516 971	\$3 516 971,2	60,72%
Total	\$446,3885	\$214,24	\$1,33	\$2 146,96	3616	57851	\$77 548 570	\$77 548 570,00	0,00%

Figure 2: Summary update of the matrix visual based on the Sales table



The screenshot displays the 'Sales' table in a Power BI model view. The table is expanded, showing a list of fields. At the top is the 'Sales' table icon and name. Below it are several measures and groups, each with an icon indicating its type (e.g., sum, count, ratio, time intelligence). The fields are organized into a hierarchical structure with expandable/collapsible arrows. Some fields have a small icon to the right, possibly indicating a relationship or a specific property.

▼ Sales
Σ Cost
> Counts
EmployeeKey
OrderDate
> Pricing
ProductKey
Profit
Profit Margin
Σ Quantity
▼ Ratios
Sales % All Region
Sales % Country
Sales % Group
ResellerKey
Σ Sales
SalesOrderNumber
SalesTerritoryKey
▼ Time Intelligence
Sales YoY Growth
Sales YTD

Figure 3: Overview of the Sales table with the additions of the measures and groups